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Review

Cryptography Based on (Idempotent) Semirings: Abandoning Tropicality?

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Abstract: This review explores the current state of public key cryptography based on idempotent semirings, with an emphasis on tropical semirings. It examines key hard problems, such as the tropical discrete logarithm problem, semidirect tropical product problem, the factorization of tropical polynomials, and the matrix power function, that underpin the security of these protocols. Given the significant number of compromised protocols based on tropical semirings, most of which are variations of the Stickel protocol, we present three algorithms and classify schemes of this type. The analysis is further illustrated with a figure that maps the relationships between tropical Stickel's-like protocols and the attacks targeting them. Additionally, the review provides an in-depth exploration of the vulnerabilities that have led to many tropical semiring-based cryptosystems being compromised. To address these challenges, the review highlights promising alternative approaches, including non-tropical idempotent platforms and non-idempotent options, such as supertropical semirings, which offer potential solutions to overcome known limitations. Furthermore, a discussion on the interplay between tropical cryptography and post-quantum cryptography is presented, raising the following question: what is the future of tropical cryptography?

Keywords: semiring; idempotent semiring; tropical cryptography; Stickel's-like protocol

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1. Introduction and Motivation

Semirings were first introduced implicitly in [1] and later appeared in other works related to the study of ideals in rings. They were further discussed in [2] in connection with the formalization of non-negative integers and rational numbers. The concept of semirings was first explicitly defined in [3], specifically in the context of the axiomatization of natural number arithmetic. Throughout time, numerous researchers have conducted extensive studies of semirings, both as a natural extension of methods from semigroup and ring theory and in connection with practical applications. These applications cover a wide range of fields, including computer science and automata theory. For example, weighted automata, which assign weights to transitions, often utilize semirings, such as the tropical semiring, to model and analyze language properties [4]. In dynamic programming, numerous algorithms, including Dijkstra's algorithm and the Viterbi algorithm, are built upon semiring structures [5]. Furthermore, specific semiring operations enable the addressing of certain NP-hard problems such as Connected Dominating Set problems and finite-domain Constraint Satisfaction Problems [6]. Tropical semirings (max, +) play a pivotal role in algorithms related to the shortest paths, minimal spanning trees, and network flows [7]. Optimization problems with particular algebraic constraints can be effectively formulated using semirings to find efficient solutions. Semirings are

widely applied in control theory and discrete event systems. For instance, tropical semirings are instrumental in modeling and analyzing discrete event systems, including automated manufacturing systems and traffic control [8]. In systems theory, semirings provide a framework for representing systems where operations like "max" and "min" replace traditional addition and multiplication, enabling the handling of nonlinearities in models [9]. They are also employed in certain signal processing tasks, such as wavelet transforms and max-plus transforms, which are used for signal filtering and compression [10]. Semirings have diverse applications in artificial intelligence (AI) and machine learning. For example, neural networks can be trained to learn semiring operations, enabling more effective reasoning in symbolic AI tasks [11]. Semiring programming offers a unified framework for integrating various AI disciplines, such as SAT solving, Bayesian inference, and convex optimization [12]. In ontology learning, semirings help structure data for machine understanding in the Semantic Web, aiding in the construction and refinement of ontologies [13]. In economics and game theory, semirings are used to model scenarios like mean payoff games, where algebraic operations over semirings describe decision-making processes involving costs and rewards [14]. In phylogenetics and biological modeling, tropical semirings are employed to model evolutionary distances and relationships between species, enabling researchers to construct accurate evolutionary trees. Additionally, semirings address optimization problems in biological systems, such as protein production pathways and cellular processes [15]. In railway scheduling and logistics, semirings provide a framework for modeling scheduling and resource allocation challenges, ensuring tasks are completed efficiently while adhering to time or resource constraints [16].

A compelling motivation for this research is the field of semiring-based cryptography, which traces its origins to the seminal work in [17], introducing one of the first cryptosystems leveraging semigroups and semirings. Since then, numerous cryptosystems have been developed using semirings, with a particular focus on tropical and idempotent semirings. These algebraic structures are being actively explored as promising alternatives to classical cryptographic frameworks, especially for constructing secure key exchange protocols. However, despite their potential, many of these systems have been compromised over time, underscoring the critical need for further research and advancements in this area.

Most of our attention will be focused on tropical cryptography due to its versatile applications in real-world scenarios that require cryptographic mechanisms, such as SSL, SSH, and IPSec. Various tropical semiring-based protocols offer exceptional robustness and efficiency, making them particularly well suited for contemporary IoT ecosystems. In these settings, devices frequently function with limited resources and demand rapid and at the same time secure communication. Furthermore, such schemes seamlessly embed into blockchains to boost the safety of trade operations and digital contracts while providing additional defenses to resist typical flaws in distributed networks. Their inherent resilience to targeted algebraic attacks also makes them perfect for protecting information in cloud-based systems, where data integrity and privacy are critical. Namely, this adaptability positions tropical algebra as a solid foundation for applying cutting-edge security protocols throughout a wide range of web-based platforms and applications.

In this work, we present the current situation in the field of cryptography based on idempotent semirings and revisit the most prominent tropical-based protocols from the literature, with a particular focus on Stickel's-like protocols, which constitute the majority of the proposed schemes. Our main contributions include analyzing the underlying problems upon which the security of these protocols relies, examining existing attacks against them, and discussing solutions and strategies for developing more secure protocols in light of the post-quantum cryptography landscape. Recognizing concerns about the future of tropical cryptography, we have also extended our discussion to cryptographic

approaches that utilize various types of semirings, including non-idempotent ones, as possible platforms.

This paper is structured as follows: Section 2 presents an overview of idempotent semirings. Section 3 discusses the current state of tropical cryptography, including existing attacks and several promising approaches to address these challenges. Section 4 explores cryptography based on idempotent but non-tropical semirings as a possible platform. Section 5 examines the potential of abandoning idempotency as a strategy for constructing more secure cryptographic schemes. Finally, Section 6 offers a discussion and concludes the paper.

2. Idempotent Semirings

Following [18], a semiring is a universal algebra, $\mathfrak{S} = (S, +, \cdot)$, defined on a nonempty set, S, with two binary operations, (+) and (·), where both $(S, +)$ and $(S, \cdot)$ are semigroups, and multiplication (·) distributes over addition (+). An element, s, of a semiring, $\mathfrak{S} = (S, +, \cdot)$, is called additively idempotent if $s + s = s$. When all elements of the semiring $\mathfrak{S}$ satisfy this property, the semiring is called additively idempotent or simply an idempotent semiring. If $(S, +)$ has a neutral element, it is called zero (0), satisfying $s + 0 = 0 + s = s$ for all $s \in S$. If $(S, \cdot)$ has a neutral element, it is called an identity (1), satisfying $s \cdot 1 = 1 \cdot s = s$ for all $s \in S$. A semiring is termed a semifield when the semigroup $(S, \cdot)$ is a group. An order, $\preccurlyeq$, in the idempotent semiring can be defined by the relation $a \preccurlyeq b$ if and only if $a + b = b$.

We recall here some of the most important examples of idempotent semirings [19,20]:

- *Interval semiring*: $\mathfrak{S} = (S, +, \cdot)$ consisting of a nonempty set, $S = [a, b]$, equipped with the operations $(\text{“}+\text{”} = \max)$ and $(\cdot = \min)$, is an idempotent semiring with $0 = a, 1 = b$, and a partial order, $\leq$.
- *Boolean algebra*: $\mathfrak{S} = (S, +, \cdot)$ including a set of two elements, $S = \{0, 1\}$, equipped with the operations $(\text{“}+\text{”} = \max, \vee)$ and $(\text{"}\cdot\text{"} = \min, \wedge)$, is an idempotent semiring with $0 = 0$ and $1 = 1$.
- $\mathfrak{S} = (S, +, \cdot)$ including a set, $S = \mathbb{R}_+$, together with the operations $(\text{“}+\text{”} = \max)$ and $(\text{"}\cdot\text{"} = \times)$(usual multiplication), is an idempotent semiring with $0 = 0$ and $1 = 1$.
- *Extended set semiring:* $\mathfrak{S} = (S, +, \cdot)$ includes a set of three elements, $S = \{0, 1, \infty\}$, such that $\infty + \infty = \infty, \infty \cdot \infty = \infty, 0 + \infty = \infty, 0 \cdot \infty = \infty, 1 + \infty = \infty$, and $1 \cdot \infty = \infty$
- *Ideal semiring*: $\mathfrak{S} = (S, +, \cdot)$ consisting of the set S of all the ideals of a ring, R, supplied with the operations inhering to the ring R, is an idempotent semiring such that $0 = 0$ (the zero ideal) and $1 = R$ (the entire ring).
- *Topological semiring*: $\mathfrak{S} = (S, +, \cdot)$ including a topology, S, on a topological space, T, equipped with the operations $\text{“}+\text{”} = \cup$ (union) and $\text{"}\cdot\text{"} = \cap$ (intersection), is an idempotent semiring with $0 = \emptyset$ and $1 = T$. A particular case of this semiring is $\mathfrak{S} = (S, \cap, \cup)$ when S is the powerset $P(T)$.
- *Language semiring*: $\mathfrak{S} = (S, +, \cdot)$ including a finite alphabet, S, equipped with the operations $\text{“}+\text{”} = \cup$ (union) and $\text{"}\cdot\text{"} =$ concatenation, is an idempotent semiring such that $0 = \emptyset$ and $1 = \boxdot$ (the empty string/word).
- *Max-min semiring:* $\mathfrak{S} = (\mathbb{R} \cup \{-\infty\} \cup \{+\infty\}, \max, \min)$, where the maximal element is $+\infty$. This semiring is both additively and multiplicatively idempotent, meaning that for any element a in the semiring, $\max(a, a) = a$ and $\min(a, a) = a$. However, it is not a semifield. For this semiring, $0 = -\infty$ and $1 = +\infty$.

- *Max-T semiring:* $\mathfrak{S} = (S, +, \cdot)$ consisting of a unit interval, $S = [0, 1]$, equipped with the operations "$+$" $=$ max and "$\cdot$"—a T-norm where $T: S^2 \to S$, is an idempotent semiring.

Remark 1. The last two semirings have been used as platforms in recent work [21]. However, the most widely used semirings for cryptographic purposes are tropical semirings, $\mathfrak{T}$, which include

$$\mathbb{R}_{max,+} = (\mathbb{R} \cup \{-\infty\}, \max, +);\ \mathbb{R}_{min,+} = (\mathbb{R} \cup \{+\infty\}, \min, +);$$

$$\mathbb{R}_{max,\times} = (\mathbb{R}_+ \cup \{0\}, \max, \times);\ \mathbb{R}_{min,\times} = (\mathbb{R}_+ \cup \{+\infty\}, \min, \times),$$

where $\mathbb{R}$ represents the field of real numbers (here, $\mathbb{R}_+ = \{x \in \mathbb{R} | x > 0\}$).

Operations in these semirings can be extended to tropical matrices, $\mathbb{R}^{m\times n} = \{M = (M_{ij}) | M_{ij} \in \mathfrak{T}, i = 1, \dots, m, j = 1, \dots, n\}$. Operations between tropical matrices are defined as follows: for matrices $M, N \in \mathbb{R}^{m\times n}$ and a real number, r, the following holds:

$$(M + N)_{ij} = m_{ij} + n_{ij}, r \cdot (M)_{ij} = r \cdot m_{ij}.$$

For two tropical matrices, $M \in \mathbb{R}^{m\times n}$ and $N \in \mathbb{R}^{n\times p}$, the matrix product is defined by

$$(M \cdot N)_{ij} = \sum_{k=1}^{n} a_{ik} \cdot b_{kj},\ i = 1, \dots, m, j = 1, \dots, p$$

Due to the non-invertibility of the additive operation (max or min) in tropical semirings, matrices in tropical algebra are non-invertible.

Tropical polynomials can be determined in the usual way:

$$P(x) = \sum_{k=0}^{n} a_k \cdot x^k,$$

where the coefficients a_k could be of any tropical semiring, $\mathfrak{T}$.

The polynomials of the matrices are also considered, and an important property is that polynomials of the same matrix commute. This property makes them suitable as building blocks for various tropical protocols, which we will explore later.

3. Cryptography Based on Tropical Semirings

As previously mentioned, tropical-based cryptography can be applied in various scenarios where cryptographic mechanisms are essential. Tropical semiring-based protocols are especially well suited for IoT setups, which have limited resources and necessitate both rapid and safe communication. Notably, the most well-known tropical-based protocols are of the Diffie–Hellman type (key exchange protocols). However, other schemes, including encryption methods like El Gamal encryption and digital signature algorithms, have also been proposed.

We discuss here the computationally hard problems that underpin the security of many tropical cryptographic protocols. Additionally, we explore the cryptographic schemes that leverage these problems and analyze the attacks developed to compromise these schemes.

We focus primarily on Stickel's-like protocols, as a significant portion of the proposed tropical-based cryptographic protocols are variations of this type, utilizing different tropical semirings. The robustness of the protocols' security is based on the computational difficulty of solving problems similar to the discrete logarithm problem (DLP), but within the framework of tropical semirings.

Other computationally hard problems that form the basis of tropical protocols' security include the semidirect product problem, the matrix power function problem, and the factorization of tropical polynomials.

3.1. Stickel's-like Protocols

We first recall the original Stickel protocol [22], since it is simple and serves as the basis for many other (tropical) protocols. The domain parameters of the protocol are a non-abelian finite group, G, and two elements, $x, y \in G$. The protocol works as follows:

(1) Alice selects two naturals, A_1 and A_2. She computes her public key, $K_A = x^{A_1} \cdot y^{A_2}$, and sends it to Bob.

(2) Bob selects two naturals, B_1 and B_2. He computes his public key, $K_B = x^{B_1} \cdot y^{B_2}$, and sends it to Alice.

(3) Alice computes $K_{AB} = x^{A_1} \cdot K_B \cdot y^{A_2}$.

(4) Bob computes $K_{BA} = x^{B_1} \cdot K_A \cdot y^{B_2}$.

At the conclusion of the protocol, Alice and Bob possess a shared secret key:

$$K_{AB} = K_{BA}$$

To break the protocol, an attacker must determine the unknown positive integers A_1 and A_2 from the equation $K_A = x^{A_1} \cdot y^{A_2}$, i.e., solve the DLP in the non-abelian finite group G. Since this is a key exchange protocol, an attacker could alternatively use Bob's public key $K_B = x^{B_1} \cdot y^{B_2}$ to compromise the shared secret.

V. Shpilrain, in [23], showed the vulnerability of the original Stickel protocol using a linear algebra attack. The same author, together with D. Grigoriev, introduced the first tropical protocol [24], which implemented the Stickel protocol over a tropical semiring, $\mathbb{R}_{min,+} = (\mathbb{R} \cup \{+\infty\}, min, +)$. The motivation for employing a tropical semiring as the platform for this protocol lies in the non-invertibility of matrices in tropical algebra, which enhances the resistance to attacks similar to those that target the original Stickel protocol. Moreover, tropical cryptographic protocols benefit from faster execution, as the underlying arithmetic operations can be performed significantly faster.

Here, we describe three main versions of the tropical-based Stickel's-like protocol. Algorithm 1 corresponds to the original Stickel protocol, Algorithm 2 represents one of the most widely used adaptations, and Algorithm 3 is the version proposed in [24].

Algorithm 1

Input:

- Tropical semiring $\mathfrak{T}$ and matrix semiring $\mathfrak{CM}$ of commuting matrices over $\mathfrak{T}$.
- Two square matrices $M, N \in \mathfrak{CM}$ of order n.
- Private numbers for Alice: (a_1, a_2) and Bob: (b_1, b_2).

Output:

- A shared secret key $K_{AB} = K_{BA}$

Steps:

1. Domain parameter setup:

- Agree on the tropical semiring $\mathfrak{T}$, the matrix semiring $\mathfrak{CM}$ of commuting matrices, and the matrices M, N.

2. Key generation by Alice:

- Select private integers (a_1, a_2)
- Compute the public key: $K_A = M^{a_1} \cdot N^{a_2}$
- Send K_A to Bob

3. Key generation by Bob:

- Select private integers (b_1, b_2)
- Compute the public key: $K_B = M^{b_1} \cdot N^{b_2}$
- Send K_B to Alice

4. Secret key determining by Alice:

- Calculate the shared key utilizing Bob's public key K_B:

$$K_{AB} = M^{a_1} \cdot K_B \cdot N^{a_2}$$

5. Secret key determining by Bob:
- Calculate the shared key utilizing Alice's public key K_A:

$$K_{BA} = M^{b_1} \cdot K_A \cdot N^{b_2}$$

6. Validation:
- Verify that $K = K_{AB} = K_{BA}$, which is guaranteed by the commutativity of the selected matrix.

To break this protocol, it is sufficient for the attacker to calculate the secret key K, exploiting the data M, N, K_A, K_B.

Algorithm 2

Input:
- Tropical semiring $\mathfrak{T}$ and matrix semiring $\mathfrak{CM}$ of commuting matrices over $\mathfrak{T}$.
- Tropical square matrix P of order n.
- Private matrices for Alice: (A_1, A_2) and Bob: (B_1, B_2). All private matrices are square matrices of order n, such that $A_1, A_2, B_1, B_2 \in \mathfrak{CM}$.

Output:
- A shared secret key $K_{AB} = K_{BA}$.

Steps:
1. Domain parameter setup:
- Agree on the tropical semiring $\mathfrak{T}$, the matrix semiring $\mathfrak{CM}$ of commuting matrices, and a matrix P.
2. Key generation by Alice:
- Select private matrices (A_1, A_2).
- Compute the public key: $K_A = A_1 \cdot P \cdot A_2$
- Send K_A to Bob.
3. Key generation by Bob:
- Select private matrices (B_1, B_2).
- Compute the public key: $K_B = B_1 \cdot P \cdot B_2$.
- Send K_B to Alice.
4. Secret key determining by Alice:
- Calculate the shared key utilizing Bob's public key K_B:

$$K_{AB} = A_1 \cdot K_B \cdot A_2.$$

5. Secret key determining by Bob:
- Calculate the shared key utilizing Alice's public key K_A:

$$K_{BA} = B_1 \cdot K_A \cdot B_2.$$

6. Validation:
- Verify that $K = K_{AB} = K_{BA}$, which is guaranteed by the commutativity of the selected matrix.

Algorithm 3

Input:
- Tropical semiring $\mathfrak{T}$ and matrix semiring $\mathfrak{M}$ over $\mathfrak{T}$.
- Three square matrices $M, N, P \in \mathfrak{M}$ of order n.
- Polynomials degree range: [1, m].
- Private tropical polynomials for Alice: (p_1, p_2) and Bob: (q_1, q_2).

Output:
- A shared secret key $K_{AB} = K_{BA}$.

Steps:
1. Domain parameter setup:

•	Agree on the tropical semiring $\mathfrak{T}$, the matrix semiring $\mathfrak{M}$, and the matrices M, N, P.
•	Set the range for polynomial degrees [1, m].
2.	Key generation by Alice:
•	Select private polynomials $p_1(x), p_2(x)$ of the degree range [1, m].
•	Compute the public key: $K_A = p_1(M) \cdot P \cdot p_2(N)$.
•	Send K_A to Bob.
3.	Key generation by Bob:
•	Select private polynomials $q_1(x), q_2(x)$ of the degree range [1, m].
•	Compute the public key: $K_B = q(M) \cdot P \cdot q_2(N)$.
•	Send K_B to Alice.
4.	Secret key determining by Alice:
•	Calculate the shared key utilizing Bob's public key K_B:
	$K_{AB} = p_1(M) \cdot K_B \cdot p_2(N)$.
5.	Secret key determining by Bob:
•	Calculate the shared key utilizing Alice's public key K_A:
	$K_{BA} = q_1(M) \cdot K_A \cdot q_2(N)$.
6.	Validation:
•	Verify that $K_{AB} = K_{BA}$, which is guaranteed by the commutativity of polynomials of the same matrix.

To break the protocols described in Algorithms 2 and 3, the attacker only needs to determine the unknown matrices X and Y from the equation

$$K_A = X \cdot P \cdot Y \tag{1}$$

(or solve a similar equation including Bob's public key, K_B).

The time complexity of Algorithm 3 is $\mathcal{O}(n^3 m)$, as it includes $\mathcal{O}(n^3)$ matrix multiplications, and for the polynomial evaluation, all the first m powers of the matrix (either M or N) must be computed, multiplied by a scalar integer, and summed together.

After the protocol in [24] was presented (which employs Algorithm 3), a number of Stickel's-like protocols appeared. These protocols are based on different tropical semirings, such as $\mathbb{R}_{max,+}$ [25,26] and both $\mathbb{R}_{max,+}$ and $\mathbb{R}_{min,+}$ [27], as well as both $\mathbb{R}_{max,\times}$ and $\mathbb{R}_{min,\times}$ [28]. In [25], several Stickel's-like protocols were developed (employing Algorithms 1–3) within the $\mathbb{R}_{max,+}$ semiring, highlighting the advantages of the max-plus semiring. In [26], the security of these protocols was enhanced by incorporating additional secret keys, the degrees of the polynomials. Furthermore, in [27,28], each user selected secret keys (polynomials) from two dual semirings. This approach requires an attacker to solve Equation (1) simultaneously in both semirings to break the protocol, thereby significantly increasing its security.

Several Stickel's-like protocols operating over the semiring $\mathbb{R}_{max,+}$ are discussed in paper [29]. Instead of using the polynomials of the matrices, the authors propose employing commuting matrices, such as Linde–de la Puente matrices (employing Algorithm 1). Additionally, the same authors suggest utilizing the quasi-polynomials of special matrices, known as Jones matrices, whose deformations, A^a and B^b, commute for any non-positive numbers, $a, b \leq 1$ (leveraging Algorithm 3).

To mitigate attacks exploiting the linearity of the powers of tropical matrices, different researchers have developed Stickel's-like protocols using circulant matrices. This approach is effective because modified tropical circulants fail to be expressed in the form of a matrix polynomial function. Examples include the protocol in [30], which employs tropical upper t-circulant matrices; the protocols in [31], which use lower s-circulant as well as anti-s-p-circulant matrices; and the protocol in [32], which suggests using Toeplitz

matrices over the max-time semiring. Notably, the schemes in [30–32] utilize the semiring of circulant matrices as the semiring $\mathfrak{CMT}$ of commuting matrices (see Algorithms 2 and 3).

Additionally, Stickel's-like protocols have been developed over various semirings. Here, we outline several examples of specially constructed idempotent semirings:

- A modified tropical structure, presented in [33], which employs Algorithm 1. In this protocol, matrices are selected from a matrix set that is the Cartesian product of tropical semirings;
- A semiring referred to as a digital semiring by the authors in [34], employing Algorithm 2.
- A protocol over tropical increasing matrices, discussed in [35], which utilizes Algorithm 2.

Furthermore, S. Alhussaini and S. Sergeev [36] proposed an implementation of Algorithm 3 over other idempotent semirings, namely the max-min and max-T semirings.

3.1.1. Attacks on Tropical Stickel's-like Schemes

The higher powers of irreducible tropical matrices exhibit near-linear periodicity. This characteristic forms the basis for many attacks on tropical Stickel's-like schemes. Specifically, an "almost linear periodic" property for sequences $\{M_i\}^{\infty}{}_{i=0}$ of the max-plus and min-plus matrices implies the existence of a period, ρ, a linear factor, c, and a defect, d, in such a way that [37]

$$M_{i+\rho} = c + M_i$$

for all $i > d$.

The first attack on the tropical Stickel's-like protocol [24] was proposed in [38]. The authors demonstrated that in the semiring $\mathbb{R}_{min,+}$, the higher powers of the tropical matrices M and N exhibit a uniform pattern: as the powers of the matrices increase, their entries begin to decrease linearly. Later, another attack on this protocol was presented in [39], improving upon the earlier approach. This refined attack suggests identifying the smallest cover rather than enumerating all possible covers.

The Stickel's-like schemes developed in [25–28] were analyzed and attacked in [40]. The authors demonstrated that in the semiring $\mathbb{R}_{max,+}$, the higher powers of the tropical matrices M and N also exhibit a uniform pattern, similarly to the behavior observed in $\mathbb{R}_{min,+}$. However, in the semiring $\mathbb{R}_{min,\times}$, while higher powers exhibit patterns, the connection between the related elements in consecutive powers follows a nonlinear model. Consequently, in the semiring $\mathbb{R}_{max,\times}$, the higher powers of matrices could not be generalized in a single expression, as different matrices displayed different patterns. Later, in [41], an attack targeting the schemes using the semirings $\mathbb{R}_{min,\times}$ and $\mathbb{R}_{max,\times}$ was presented. This attack identified an "almost multiplicatively periodic" property in the sequences $\{M_i\}^{\infty}{}_{i=0}$ of the max-time and min-time matrices. Specifically, this property suggests the existence of a period, p, a factor, c, and a defect, d, such that [41]

$$M_{i+p} = c \cdot M_i$$

for all $i > d$.

As noted earlier, several works have utilized circulant matrices to construct their Stickel's-like protocols. Special heuristic attacks targeting such protocols, such as those presented in [30–32], were proposed in [42]. Moreover, the authors argued that their method achieved "a high success rate" when used to attack the original tropical Stickel's-like protocol [24].

An attack on the modified structure proposed in [33] was presented by the authors of [42]. This attack involves solving the tropical two-sided discrete logarithm with a

"shift" problem. Another attack on the key exchange protocol, which utilizes a modified tropical structure, was examined in [43]. This attack exploits the periodicity of the difference between the pseudo key and the actual key. Its advantage lies in the ability to extract the common secret key using only publicly available parameters.

Recently, in [44], Alhussaini and Sergeev introduced new methods for attacking tropical Stickel's-like schemes using mixed-integer linear programming. By applying this technique, they demonstrated the insecurity of their Stickel-like protocol [36], built on the max-time semiring, as well as the protocol from [34], the security of which is based on the Three-Satisfiability (3-SAT) problem.

Figure 1 illustrates various Stickel's-like schemes along with the main attacks targeting them.

I prefer to be: In fact, solving Equation (1) under the condition $X \otimes Y = C$ can be interpreted as determining the matrices X and Y from the system [38].

$$\begin{cases} M \cdot X = X \cdot Z, \\ Y \cdot N = N \cdot Y, \\ X \cdot Y = E \end{cases} \tag{2}$$

(or solve a similar system including Bob's public key, K_B). A universal attack on this system applicable to any idempotent semiring was presented in [45].

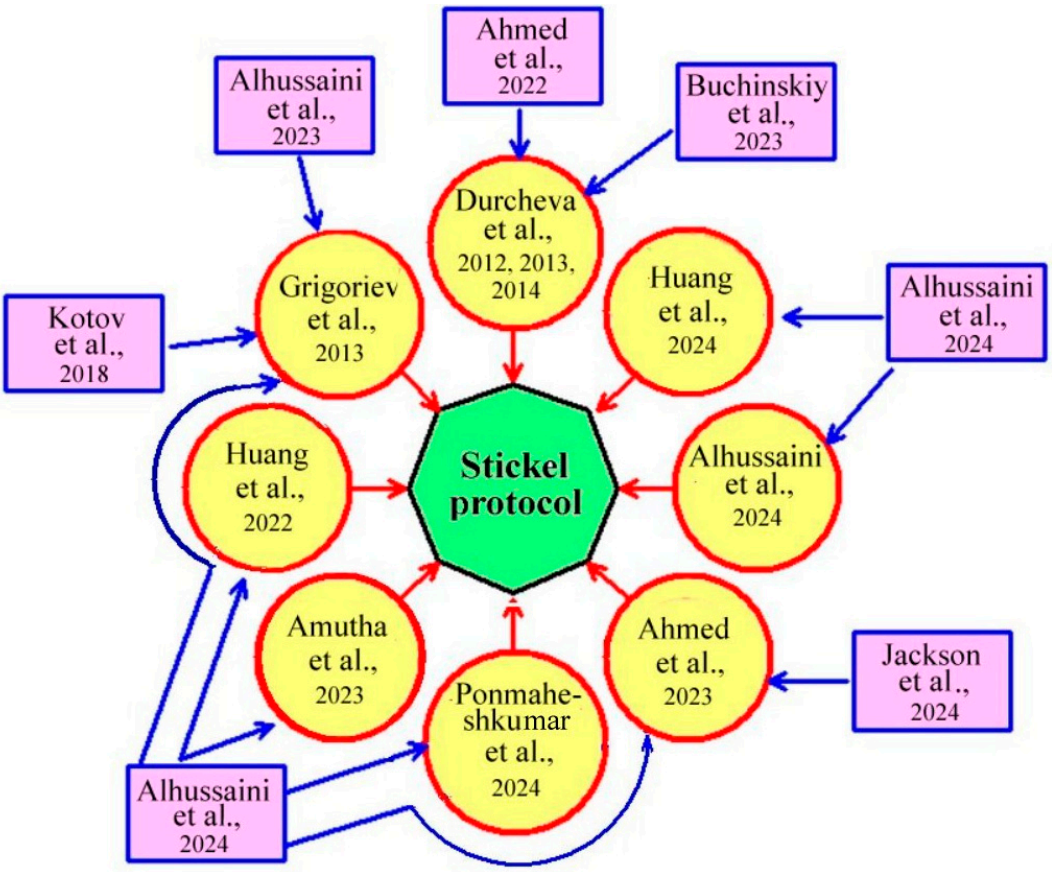

Figure 1. Tropical Stickel's-like protocols [24–28,30–34,36] and attacks on them [38–44].

3.1.2. The Future of Stickel's-like Schemes

Although most known tropical Stickel-like protocols have been compromised, promising approaches are emerging for developing efficient and secure protocols inspired by Stickel's original concept.

Approach 1: Employing new tropical (idempotent) semirings as platforms for such protocols. For instance, the tropical Stickel's-like protocol based on Linde–de la Puente

matrices has demonstrated resilience against known attacks, making these matrices suitable building blocks for secure tropical key exchange protocols (see [21]). Another promising direction, also presented in the same paper, involves utilizing non-tropical yet idempotent semirings constructed from various classes of T-norms.

Additional intriguing possibilities for enhancing the tropical Stickel protocol are presented in recent protocols proposed by Jackson and Perumal. One approach employs a tropical triad semiring [46], while the other utilizes Collatz codes [47]. Both protocols effectively address key vulnerabilities typically associated with tropical schemes, including linear algebra attacks and the Kotov–Ushakov (KU) attack [38]. The same authors also suggested another approach in [48]: the use of tropical determinants. This method has the added advantage of mitigating linear algebra attacks as well.

Approach 2: Moving beyond idempotent semirings to explore alternative semirings as potential platforms. This approach opens new possibilities for constructing secure protocols and will be further discussed in Section 5.

3.2. Protocols Based on a Semidirect Product

Another interesting class of tropical protocols is based on semidirect products. A basic key exchange protocol using a semidirect product was first presented in [49]. Grigoriev and Shpilrain [50] adapted this protocol for tropical semirings. According to [50], in a tropical semiring, the semidirected product can be defined as follows: Let G be a semigroup and $\mathfrak{T}$ be a tropical semiring. The semidirect product of $\mathfrak{T}$ and G is a set, $\mathfrak{T} \rtimes G = \{(t, g) |\, t \in \mathfrak{T}, g \in G\}$, with the semigroup operation

$$(t_1, g_1) \cdot (t_2, g_2) = \left(t_1^{g_2} \cdot t_2, g_1 \cdot g_2\right).$$

In the protocol in [48], the authors used the semigroup of $\mathbb{Z}$ with respect to the operation $x \circ y = x + y + xy$ (for $x \circ y \in \mathbb{Z}$) and constructed a semidirect product as the set $\mathfrak{T} \rtimes \mathbb{Z} = \{(t, x) |\, t \in \mathfrak{T}, x \in \mathbb{Z}\}$, equipped with the semigroup operation

$$(t_1, x_1)(t_2, x_2) = ((t_1 \circ x_2) + t_2, x_1 \circ x_2)$$

The protocol works as follows:

Domain parameters: The tropical matrix semiring $\mathcal{M} = (M_n(\mathbb{Z}), \oplus, \otimes)$ and two matrices, M and H, chosen from $\mathcal{M}$.

Protocol steps:

(1) Alice picks a positive integer, m, as her private key. She computes $(M, H)^m = (K_A, H^m)$ and transmits K_A to Bob.
(2) Bob picks a positive integer, n, as his private key. He computes $(M, H)^n = (K_B, H^n)$ and transmits K_B to Alice.
(3) Alice calculates the shared secret key $K_{AB} = (K_B \circ H^m) \oplus K_A$.
(4) Bob calculates the shared secret key $K_{BA} = (K_A \circ H^n) \oplus K_B$.

By the end of the protocol, both users will have the same shared secret key, $K_{AB} = K_{BA}$, as the common key is equal to the first component of the expression $(M, H)^{m+n}$.

3.2.1. Attacks on Schemes Using Semidirect Products

Four different approaches to attacking one of the protocols introduced in [50] have been proposed (see [37,51–53]). Two of these attacks, namely [37,51], successfully recovered the private exponent by examining the sequence of exchanged elements. Isaac and Kahrobaei [37] utilized the characteristic of the matrix sequence exhibiting "almost linear periodicity". Rudy and Monico [51] took advantage of the fact that the sequences of matrices in the protocol are monotonically decreasing. This allowed for the application of a binary search technique to efficiently recover the private exponent. Muanalifah and Sergeev in [52] suggested an attack that relies on solving the tropical discrete logarithm

problem, which is central to breaking the protocol. Another attack, presented in [53], utilized the property that a difference between the secret key and the product of the shared matrices remains independent of the users' selection of public data. As a result, this difference becomes independent of the higher powers of the public matrix H, maintaining only particular constant elements dictated by the defect and period of the matrices M and H.

3.2.2. The Future of Schemes Based on Semidirect Products

The current state of semidirect product key exchange protocols is analyzed in [54]. The authors argue that, at present, only the Dihedral Hidden Subgroup Problem (DHSP) is vulnerable to quantum attacks. However, as far as we are aware, there is only one protocol based on a tropical semidirect product, which has already been successfully attacked. This raises significant doubts about the future viability of protocols of this type.

3.3. Protocols Using Matrix Power Function (MPF)

A basic protocol using the matrix power function was introduced in [55]. The concept of the MPF is as follows: Let $Q = (q_{ij})_{n\times n}$ be a matrix that when powered by another matrix, $Y = (y_{ij})_{n\times n}$, from the right, yields a resulting matrix, $P = (p_{ij})_{n\times n}$ (denoted $P = Q^Y$). Similarly, when matrix Q is powered by a matrix, $X = (x_{ij})_{n\times n}$, from the left, it yields a matrix, $S = (s_{ij})_{n\times n}$ (denoted $S = {}^X Q$). The elements of the resulting matrices are computed as follows:

- For P, $p_{ij} = \prod_{k=1}^{n} q_{ik}{}^{y_{kj}}$;
- For S, $s_{ij} = \prod_{k=1}^{n} q_{kj}{}^{x_{ik}}$.

When the elements of the base matrix Q are drawn from a (semi)group, $\mathfrak{G}$, and the elements of the matrices X and Y are taken from the ring $\mathbb{Z}_m$ (m denotes the maximum multiplicative order of the elements in $\mathfrak{G}$), the matrix power function (MPF) is formally defined as a mapping [55]:

$$\text{Mat}\,(\mathbb{Z}_m)\times\ \text{Mat}(\mathfrak{G}) \rightarrow \text{Mat}(\mathfrak{G}),$$

Alternatively,

$$\text{Mat}(\mathfrak{G})\ \times\ \text{Mat}\,(\mathbb{Z}_m)\rightarrow \text{Mat}(\mathfrak{G}),$$

which will be referred to as $S = {}^X Q$, $(P = Q^Y)$.

This definition provides the foundation for applications in various algebraic structures. Durcheva adapted the protocol in [55], proposing the tropical semiring as a platform [56]. As highlighted above, some of the developed tropical protocols include encryption schemes, such as the one proposed in [56]. The presented scheme incorporates a double key exchange mechanism.

The MPF key exchange protocol of the first phase operates as follows: The domain parameters include a tropical semiring, $\mathfrak{T}$, which can be either $\mathbb{R}_{max,+}$ or $\mathbb{R}_{min,+}$, a set of circulant matrices, $\mathcal{C}$, over $\mathfrak{T}$, and two matrices, $C_1, C_2 \in \mathcal{C}$, of the order n. Additionally, a random matrix, M, from a matrix semigroup, $\mathcal{G}$, is selected.

(1) Alice picks as a private key two circulant matrices, $A_1, A_2 \in \mathcal{C}$. She calculates

$$K_A = {}^{A_1}C_1 \otimes {}^{A_2}C_2 \otimes M$$

and transmits K_A to Bob.

(2) Bob picks as a private key two circulant matrices, $B_1, B_2 \in \mathcal{C}$. He calculates

$$K_B = {}^{B_1}C_1 \otimes {}^{B_2}C_2 \otimes M$$

and transmits K_B to Alice.

(3) Alice determines her secret key in the following manner:

$$K_{AB} = {}^{A_1}C_1 \otimes {}^{A_2}C_2 \otimes K_B.$$

(4) Bob determines his secret key in the following manner:

$$K_{BA} = {}^{B_1}C_1 \otimes {}^{B_2}C_2 \otimes K_A.$$

At the conclusion of the protocol, both parties obtain the same secret key, thanks to the commutative properties of circular matrices.

To compromise the protocol, it is sufficient to solve the following problem.

Provided with a matrix, K_A, the circulant matrices C_1 and C_2, and a matrix, M, determine two circulant matrices, X and Y, such that

$$K_A = {}^{X}C_1 \otimes {}^{Y}C_2 \otimes M.$$

The common secret key $K_{AB} = K_{BA}$, obtained in the first phase, serves as the input matrix P for the second phase, which implements a Stickel's-like protocol (Algorithm 3, Section 3.1). The resulting shared secret key, a tropical matrix, F, is then used for encryption/decryption purposes. To encrypt a message, S, the sender computes $C = F \oplus S$, where $\oplus$ denotes the bitwise sum modulo 2 applied to all elements of the matrices F and S. To decrypt the ciphertext C, the receiver calculates $S = F \oplus C$, which gives the following:

$$F \oplus C = F \oplus F \oplus S = S.$$

3.3.1. Attacks on Protocols Employing MPF

The original protocol introduced in [55] was successfully attacked using linear algebra techniques, as demonstrated in [57]. In response, the authors proposed alternative platforms where the associated decisional problem is NP-complete [58].

The tropical protocol suggested in [56] was later attacked in [59] by solving the resulting linear equations within the tropical semiring.

3.3.2. The Future of the Schemes Based on the MPF

In the non-commuting platform group, the discrete logarithm mapping is inapplicable, leading to the algebraic analysis of the associated decisional problem being classified as NP-complete [58]. On the one hand, Mihalkovich, Zitkevicius, and Sakalauskas, in [60], analyzed the protocols based on the MPF over a tropical semiring and concluded that they were insecure. One the other hand, a promising platform for the MPF's action could be a supertropical semiring, as demonstrated in [61].

3.4. Protocols Based on Factorization of Tropical Polynomials

The problem of tropical polynomial factorization can be stated as follows: given a tropical polynomial, r(x), determine two tropical polynomials, p(x) and q(x), such that $r(x) = p(x) \otimes q(x)$.

Under several conditions, the problem of factoring polynomials over a tropical semiring has been proven to be NP-complete (Th 7, [62]).

A digital signature scheme whose security is based on this problem was first introduced in [63]. The protocol operates in the tropical (min-plus) algebra $\mathcal{T}$ of polynomials over the ring of integers $\mathbb{Z}$. The private parameters consist of two tropical polynomials, P and Q, of the degree deg, whose coefficients are confined to the interval $[0, r]$, where r is a domain parameter. Here, the public parameters include the polynomial $R = P \otimes Q$ and a hash function, H. Users mutually agree on a procedure for converting the values of H into one-variable tropical polynomials.

To sign the message m, the sender needs to carry out the following:

(1) Convert the hashed message to a tropical polynomial, M.
(2) Select as their private key two random tropical polynomials, A_1 and A_2, of the degree deg, with all coefficients in the range $[0, r]$, and compute $A = A_1 \otimes A_2$.
(3) The signature is the triple of the polynomials $(M \otimes P \otimes A_1, M \otimes Q \otimes A_2, A)$.

To verify the message, the verifier performs the following steps:

(1) Calculates the hash $H(m)$ and transforms $H(m)$ into a tropical polynomial, M, of the degree deg.
(2) Checks that the polynomials $M \otimes P \otimes A_1$ and $M \otimes Q \otimes A_2$ are of the degree $3deg$ and that the degree of the polynomial A is equal to $2deg$. If not, the signature will be rejected. Checks that neither $M \otimes P \otimes A_1$ nor $M \otimes Q \otimes A_2$ is a multiple of $M \otimes R$ or $M \otimes A$. If either is, the signature is rejected. Additionally, verifies that the coefficients in the polynomials $M \otimes P \otimes A_1$ and $M \otimes Q \otimes A_2$ are of the interval $[0, 3r]$ and, simultaneously, that the coefficients in the polynomial A are of the interval $[0, 2r]$. Otherwise, the signature will be invalid.
(3) The verifier calculates $S = (M \otimes P \otimes A_1) \otimes (M \otimes Q \otimes A_2)$. The signature is accepted if and only if $S = M \otimes M \otimes R \otimes A$.
(4) The protocol is correct (see [64]), since $S = (M \otimes P \otimes A_1) \otimes (M \otimes Q \otimes A_2) = M \otimes M \otimes (P \otimes Q) \otimes (A_1 \otimes A_2) = M \otimes M \otimes R \otimes A$.

3.4.1. Attacks on Protocols Based on Factorization of Tropical Polynomials

The protocol from [63] was attacked in [64] through two different methods. The first method involved searching for small modifications to the coefficients of the resultant matrices, while the second suggested finding only small-degree divisors instead of performing a full factorization.

The main tactic of the attack, outlined in [65], was to divide one public polynomial by another. Practically, it involved techniques such as swapping specific symbols in the signature computation, factoring the hash, factoring the public key, and re-hashing a valid signature.

3.4.2. The Future of the Protocols Utilizing the Factorization of Tropical Polynomials

To avoid the attacks described in [64,*65*], the approach proposed in [66] suggests using the Fiat–Shamir Transform (FST), asserting that this method could be extended to other cryptosystems as well.

In response to recent attacks, the authors of the protocol in [64] revised their protocol and released an updated version [67]. In this improved version, an additional parameter was added to the secret key, chosen by the sender (an extra tropical polynomial), thereby expanding the signature to a sextuple of polynomials instead of the original triple. The authors also incorporated polynomial addition to counter the division attack proposed by Brown and Monico in [65].

4. Other Protocols Using Idempotent Semirings as a Platform

In this section, we examine several recently proposed protocols that leverage various idempotent semirings. The security of these protocols is based on problems distinct from those discussed earlier.

4.1. Protocols Utilizing the Problem of Multiple Exponentiations of Tropical Matrices

This category of protocols relies on the computational difficulty of determining multiple exponentiations of tropical matrices. Protocols based on this problem were proposed by [68] and later extended in [69]. The authors proposed to reduce the multiple exponentiations of tropical matrices (METM) problem to a constructive membership problem in a semigroup, achievable within polynomial time. This problem is considered

computationally difficult and resistant to quantum computer attacks, as demonstrated in [70]. Furthermore, Huang et al. [71] argued that it has not been proven that the METM problem can be converted to the Hidden Subgroup Problem, which is known to be vulnerable to quantum cryptanalysis.

4.2. Protocols Employing Tropical Block Matrices

A protocol utilizing tropical block matrices as building blocks was proposed by Durcheva and Danilchenko [72]. The matrix decomposition problem for block matrices underpins the security of this protocol. An alternative approach to compromising the protocol is to attempt to decompose the matrix into the component block matrices. For matrices of the size n, the leading algorithms for this decomposition, such as Gauss–Jordan elimination and LU decomposition, demonstrate exponential time complexity ($O(2^n)$). Given the computational requirements, the problem becomes infeasible for practical use, especially with the parameter data suggested by the authors. While a recent attack on this protocol in certain scenarios, as discussed in [45], highlights potential vulnerabilities, we believe that the use of tropical block matrices as a cryptographic primitive remains a promising approach.

4.3. The Encryption Protocol Based on Max-Plus (MP) Wavelet Transforms

MP wavelets are computationally simple and efficient, as highlighted by [73]. A key advantage of using MP wavelets in encryption schemes is that they provide an extremely large number of potential decryption keys, even for short plaintexts. Another encryption scheme applying an integer wavelet transform that utilizes the min-max-plus lifting scheme was introduced in [74].

4.4. Protocols Based on T-Norms

The authors of several Stickel's-like protocols based on max-min (max-T) semirings [36] argued that these protocols offer an advantage over tropical ones due to their greater resilience against two-sided discrete logarithm attacks. Notably, existing attack methods, such as the Kotov–Ushakov approach, have proven to be relatively ineffective.

4.5. Protocols Using Tropical Congruent Transformation

A public key encryption and key exchange protocol that utilizes the congruent transformation of a symmetric tropical matrix by a tropical circular matrix was presented in [75]. The authors argue that the proposed cryptosystems can withstand all known attacks relevant to tropical cryptography. This robustness is attributed to the implemented commutative semiring, which is not representable by public matrices, and the fact that matrix addition and matrix exponentiation operations are not utilized in these cryptosystems.

5. Abandoning Idempotency: A Path Toward Building More Secure Schemes?

Exploring alternative (non-idempotent) algebraic structures suggests new opportunities for constructing secure cryptographic protocols. Non-idempotent semirings, with their more complex algebraic properties, may provide enhanced resistance to known attacks that exploit idempotency. This section discusses the potential benefits of moving beyond idempotent frameworks and examines various proposals leveraging non-idempotent semirings as a platform for cryptographic schemes. Here, we present two different approaches for employing non-idempotent semirings.

5.1. Cryptography Based on Semigroup Action Problem (SAP) in Simple Semirings

A pioneering cryptosystem based on semigroups and semirings was presented in [17] by Maze, Monico, and Rosenthal. The platform semiring they employed was a finite congruence-simple semiring, chosen specifically to mitigate the risk of Pohlig–Hellman-type reduction attacks [76]. A cryptosystem based on semigroups and semirings was also developed by Trendafilov and Durcheva [77]. Atani contributed to this field with two cryptosystems [78,79] utilizing semi-modules over factor semirings. Additionally, Dwivedi et al. offered a scheme [80] built upon polynomials over noncommutative factor semirings. These protocols primarily rely on the semigroup action problem (SAP) as their foundational computational challenge.

In the protocol proposed in [17], a finite semiring, R, with a center, C, is utilized. The center C consists of elements of R that commute with every other element in R. Let $C[M]$ denote the commutative semiring formed by the polynomials of the matrices $M \in \mathrm{Mat}_n(R)$, with coefficients from C. Given two matrices, M_1 and M_2, the protocol defines a linear action as follows [17]:

$$(C[M_1] \times C[M_2]) \times \mathrm{Mat}_n(R) \to \mathrm{Mat}_n(R),$$

$$((p(M_1), q(M_2)), X) \mapsto p(M_1) \cdot X \cdot q(M_2).$$

The security of the protocol relies on the problem called the two-sided action problem of matrix semirings.

Provided with three matrices, $M_1, M_2, Q \in Mat_n(R)$, and a matrix, $A \in C[M_1] \cdot Q \cdot C[M_2]$, find the matrices $B_1 \in C[M_1]$ and $B_2 \in C[M_2]$ such that $A = B_1 \cdot Q \cdot B_2$.

However, SAP-based protocols have been successfully attacked by various methods, indicating potential vulnerabilities in their underlying security assumptions. For example, in [81], a heuristic algorithm was proposed that solves the two-sided action problem in most cases; Gnilke also proposed several attacks in [82].

What about the future of SAP-based cryptography?

In a recent paper [71], Huang, Peng, and Deng conducted a thorough cryptanalysis of SAP-based protocols. They showed that the majority of cryptosystems grounded in the SAP on a module are vulnerable to quantum cryptanalysis. While the SAP on a semi-module is NP-hard, its quantum computational complexity is exponential in specific cases. The attacks described above exploit the structure of semigroups, rendering most SAP implementations vulnerable to being compromised. Despite these challenges, cryptographic group actions, a closely related concept, have gained prominence in isogeny-based cryptography [83]. The future of SAP-based cryptography remains uncertain, with ongoing research focused on exploring its potential applications while addressing critical security concerns.

The findings highlight that while SAP-based protocols may offer theoretical security against classical attacks, their resilience in a post-quantum context is limited, particularly when relying on modules. Semi-modules, due to their increased complexity, might still hold promise under careful parameter selection.

5.2. Cryptography over Supertropical Semiring

The supertropical semiring, which extends the max-plus semiring, was first introduced by Izhakian and Rowen in their foundational work [84]. This semiring retains the beneficial properties of tropical semirings while incorporating additional elements that enhance its potential for cryptographic applications.

It should be noted that the security of protocols that employ supertropical semirings relies on two critical factors: the NP-hardness of the solving systems of nonlinear equations in the supertropical setting and their resistance to linear algebra attacks due to the lack of an additive inverse in supertropical semirings. As discussed above, a protocol

utilizing a supertropical semiring has emerged [61], presenting a promising platform for the development of future cryptosystems. Another recent protocol, which discusses a multi-party key exchange scheme based on the supertropical semiring, was presented in [85].

6. Discussion, Conclusions, and Prospects

6.1. Discussion

Classical asymmetric cryptography algorithms, such as Diffie–Hellman key exchange (DHKE) and RSA, are based on several well-known NP-hard problems. For example, DHKE relies on the discrete logarithm problem (DLP); similarly, a number of tropical-based cryptosystems are built on analogous problems but within the framework of tropical semirings. Other NP-hard problems that underpin various tropical-based cryptographic schemes include the tropical semigroup action problem, the tropical semidirect product problem, Solving Two-Sided Linear Systems in Tropical Semirings, the factorization of tropical polynomials, the tropical matrix power function problem, and the Three-Satisfiability (3-SAT) problem.

Post-quantum cryptography, on the other hand, relies on NP-hard problems that are considered more challenging than those used in classical cryptography and are designed to withstand quantum attacks. According to NIST's Post-Quantum Cryptography Project [86], the key categories of post-quantum cryptography at present encompass [87]; Hash-Based Cryptography (use case: digital signatures) [88]; Lattice-Based Cryptography (use case: digital signatures and encryption/key encapsulation) [89]; Code-Based Cryptography (use case: digital signatures and encryption/key encapsulation) [90]; Multivariate-Based Cryptography (use case: digital signatures and encryption/key encapsulation) [91]; and isogeny-based cryptography (use case: digital signatures) [92].

Unfortunately, tropical-based cryptography has not yet been included in this list.

Yet, at the end, we want to highlight a new trend in semiring-based cryptography—namely, its application to IoT devices. In this context, it is important to mention protocols such as those described in [48,61].

6.2. Conclusions and Prospects

This review underscores the pivotal role that idempotent semirings, particularly tropical semirings, play as platforms for constructing secure cryptographic protocols. A significant portion of our analysis focused on Stickel's-like protocols, identifying their underlying security problems and examining their resilience to various attacks. However, our findings highlight that many established tropical protocols remain vulnerable, prompting a shift in cryptographic research toward alternative platforms.

In response to these vulnerabilities, cryptographers have begun investigating both idempotent and non-idempotent semirings as potential platforms for enhancing security. This review examined two key non-idempotent approaches:

- The SAP-based approach. While pioneering in its time, protocols based on the semigroup action problem (SAP) in various non-idempotent semirings have demonstrated susceptibility to heuristic and quantum attacks, limiting their viability in modern cryptographic applications.
- The supertropical semiring-based approach. A novel and promising platform, the supertropical semiring extends tropical semirings by introducing additional elements (e.g., ghost elements) that provide a richer algebraic structure and increased complexity. The NP-hardness of solving nonlinear equations over the supertropical semiring further underscores its potential for cryptographic use. Notably, the

absence of significant known attacks against protocols utilizing this semiring supports its viability as a secure alternative.

Future work should focus on further developing and analyzing non-tropical/non-idempotent protocols, ensuring their robustness in the face of evolving cryptographic challenges.

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